HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Operated and tested electro-mechanical assemblies using **test equipment**, performing fault isolation on automated systems to identify root causes of performance deviations.
- Retrieved and analysed engineering data, including **yield performance results** and operational metrics, to support process improvement initiatives.
- Documented test results and prepared comprehensive written reports and pass-down notes for engineering teams, ensuring clear communication of findings.
- Supported **cross-functional equipment deployment teams**, collaborating with outside suppliers to ensure equipment met specifications and timelines.
- Assembled and tested equipment fixtures and electronics under the supervision of project engineers, contributing to engineering studies and validations.
- Followed and enforced **Dexcom's Safety and Lock Out/Tag Out (LOTO) program**, ensuring all activities were performed with appropriate safety requirements.
- Updated equipment-related documentation, including manuals, drawings, and specifications, to reflect current operational status and configurations.

SKILLS

Core Competencies – Problem Solving, Data Analysis, Technical Documentation, Process Improvement, Hands-on Testing, Communication, Technical Writing, Equipment Assembly, Fault Isolation, Safety Procedures, Lean Manufacturing Concepts

Analysis Tools – Minitab, Tableau, SQL

Methodologies – A3, 5 Whys, IQs, OQs

Domain Knowledge – Electro-mechanical Assemblies, Robotic Equipment, Vision Systems, Automated Systems, High Volume Manufacturing Equipment

Operating Systems – Linux, Windows

PROJECTS

Automated Test Fixture for Electro-Mechanical Assemblies

- Designed and built an automated test fixture to validate electro-mechanical assemblies, integrating **Python scripts** for control and data logging.
- Developed SQL queries to extract and analyse test results from a PostgreSQL database, identifying failure patterns in **~1,500 units** tested.
- Authored comprehensive technical documentation, including assembly guides and operational manuals, for the new testing equipment.

Yield Improvement Analysis for Production Line

- Analysed **production line yield data** using Minitab and **Tableau dashboards** to identify key factors contributing to performance variations.
- Applied **5 Whys methodology** to investigate root causes for recurring defects identified in the data, proposing targeted corrective actions.
- Collaborated with manufacturing engineers to implement process changes, contributing to a measured **12% reduction in yield loss** over a 3-month period.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)